

Lipid-modifying effects of krill oil in humans: systematic review and meta-analysis of randomized controlled trials

Sorin Ursoniu, Amirhossein Sahebkar, Maria-Corina Serban, Diana Antal, Dimitri P. Mikhailidis, Arrigo Cicero, Vasilios Athyros, Manfredi Rizzo, Jacek Rysz, Maciej Banach; for the Lipid and Blood Pressure Meta-analysis Collaboration Group

Context: Some experimental and clinical trials have shown that krill oil, extracted from small red crustaceans, might be an effective lipid-modifying agent, but the evidence is not conclusive. **Objective:** The effect of krill oil supplements on plasma lipid concentrations was assessed through a systematic review of the literature and a meta-analysis of available randomized controlled trials. **Data sources:** PubMed and Scopus were searched up to March 25, 2016, to identify RCTs investigating the effect of krill oil supplements on plasma lipids. **Study selection:** Randomized controlled trials that investigated the impact of at least 2 weeks of supplementation with krill oil on plasma/serum concentrations of at least one of the main lipid parameters (ie, total cholesterol, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, or triglycerides) and that reported sufficient information on plasma/serum lipid levels at baseline and at the end of study in both krill oil and control groups were eligible for inclusion. **Data extraction:** Two reviewers independently extracted the following data: first author's name, year of publication, study location, study design, number of participants in the krill oil and control groups, dosage of krill oil, type of control allocation, treatment duration, demographic characteristics of study participants, and baseline and follow-up plasma concentrations of lipids. Effect size was expressed as the weighted mean difference (WMD) and 95% confidence interval (95%CI). **Results:** Meta-analysis of data from 7 eligible trials (14 treatment arms) with 662 participants showed a significant reduction in plasma concentrations of low-density lipoprotein cholesterol (WMD, -15.52 mg/dL; 95%CI, -28.43 to -2.61 ; $P=0.018$) and triglycerides (WMD, -14.03 mg/dL; 95%CI, -21.38 to -6.67 ; $P<0.001$) following supplementation with krill oil. A significant elevation in plasma concentrations of high-density lipoprotein cholesterol was also observed (WMD, 6.65 mg/dL; 95%CI, 2.30 to 10.99 ; $P=0.003$), while a reduction in plasma concentrations of total cholesterol did not

Affiliation: S. Ursoniu is with the Department of Functional Sciences, Discipline of Public Health, Victor Babes University of Medicine and Pharmacy, Timisoara, Romania. A. Sahebkar is with the Biotechnology Research Center, Mashhad University of Medical Sciences, Mashhad, Iran. M.-C. Serban is with the Department of Functional Sciences, Discipline of Pathophysiology, Victor Babes University of Medicine and Pharmacy, Timisoara, Romania. D. Antal is with the Discipline of Pharmaceutical Botany, Faculty of Pharmacy, Victor Babes University of Medicine and Pharmacy, Timisoara, Romania. D.P. Mikhailidis is with the Department of Clinical Biochemistry, Royal Free Campus, University College London Medical School, University College London, London, United Kingdom. A. Cicero is with the Medical & Surgical Sciences Department, Alma Mater Studiorum – University of Bologna, Bologna, Italy. V. Athyros is with the Second Propedeutic Department of Internal Medicine, Medical School, Aristotle University of Thessaloniki, Hippocraton General Hospital, Thessaloniki, Greece. M. Rizzo is with the Biomedical Department of Internal Medicine and Medical Specialties, University of Palermo, Palermo, Italy. J. Rysz and M. Banach are with the Department of Hypertension, Medical University of Lodz, Lodz, Poland.

Correspondence: M. Banach, Department of Hypertension, WAM University Hospital, Medical University of Lodz, Zeromskiego 113, 90-549 Lodz, Poland. Email: maciejbanach@aol.co.uk. Phone: +48-42-639-37-71.

Key words: Euphausiacea krill, krill oil, meta-analysis, plasma lipids.

©The Author(s) 2017. Published by Oxford University Press on behalf of the International Life Sciences Institute. All rights reserved. For Permissions, please e-mail: journals.permissions@oup.com.

reach statistical significance (WMD, -7.50 mg/dL; 95%CI, -17.94 to 2.93 ; $P = 0.159$). **Conclusion:** Krill oil supplementation can reduce low-density lipoprotein cholesterol and triglycerides. Additional clinical studies with more participants are needed to assess the impact of krill oil supplementation on other indices of cardiometabolic risk and on the risk of cardiovascular outcomes.

INTRODUCTION

Krill are red shrimp-like crustaceans that live in the cold waters of the southern (Antarctic) and northern (Arctic) polar seas.¹ Although krill is the main food source for whales, it remains an abundant biomass on earth because of its high proliferation properties.² Krill have a short life-span (1–2 years) and, because they live in clean waters, are free of heavy metals, pesticides, and dioxins.² The major krill species, *Euphausia superba*, known as Antarctic krill, is the source of extracted krill oil.³ Many experimental and clinical trials have reported a protective role of krill oil in chronic inflammation,² dyslipidemia,⁴ premenstrual syndrome,⁵ osteosarcoma,⁶ and rheumatoid arthritis.⁷ Krill oil is rich in phospholipids that contain n-3 polyunsaturated fatty acids, mainly eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA).⁸ It also contains various potent antioxidants, such as the carotenoid astaxanthin, and vitamins A and E.⁹ The association between phospholipids and long-chain n-3 fatty acids greatly facilitates the passage of fatty acid molecules through the intestinal wall, thereby increasing their bioavailability and improving the ratio of n-3 to n-6 fatty acids.^{3,10}

Most of krill oil's benefits have been attributed to its high EPA and DHA content.¹¹ The well-recognized metabolic effects of EPA and DHA include lowering of triglyceride (TG) and very low-density lipoprotein cholesterol levels.¹² Choline, an ingredient in krill oil, reduces homocysteine, transports different lipids, and is involved in the synthesis of neurotransmitters (acetylcholine) and phospholipids (sphingomyelin, lyso-phosphatidylcholine, phosphatidylcholine, and choline plasmalogen).¹³

The lipid-modifying effects of krill oil have been demonstrated in some trials, but the data are still inconclusive. Therefore, all published trials on krill oil supplementation were reviewed systematically and assessed to determine the overall efficacy of krill oil on plasma lipids using meta-analysis.

METHODS

Search strategy

This systematic review and meta-analysis was performed in compliance with the PRISMA guidelines.¹⁴

The Scopus and MEDLINE databases were searched using the following search terms in titles and abstracts (also in combination with MeSH terms): (“randomized controlled trial” OR “randomized” OR “placebo” OR “cholesterol” OR “triglyceride” OR “LDL” OR “LDL-C” OR “LDL-cholesterol” OR “HDL” OR “HDL-C” OR “HDL-cholesterol” OR “hyperlipidemia” OR “hyperlipidemic” OR “hypolipidemic” OR “dyslipidemia” OR “dyslipidemic”) and (“krill oil” OR “euphausiacea” OR “euphausia superba”). The wild-card term “*” was used to increase the sensitivity of the search strategy. The literature search was limited to papers published in English and to studies conducted in humans. The literature was searched from inception to March 25, 2016.

Two investigators (S.U. and M-C.S.) independently assessed each article, carried out data extraction, and performed quality assessment. Disagreements were resolved by consensus and discussion with a third party (M.B.).

The PICOS (Participants, Intervention/exposure, Comparison, Outcomes, Study design) criteria used to define the research question are shown in Table 1.

Study selection

The following criteria were used for the inclusion of original articles in the meta-analysis: a randomized controlled design; investigation of the effects of krill oil on plasma/serum concentrations of at least one of the main lipid parameters (ie, total cholesterol, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), or TGs; measurement of either plasma/serum concentrations of lipids at baseline and at the end of study in both krill oil and control groups or the net change in lipid levels during the study; and administration of krill oil for a period of at least 2 weeks. The following exclusion criteria were applied: experimental studies; uncontrolled trials; administration of krill oil preparations for <2 weeks; and lack of reporting of baseline or follow-up lipid concentrations or the net change in lipid levels during the study.

Data extraction

Studies meeting the inclusion criteria were reviewed, and data regarding authors, study location, publication

Table 1 PICOS criteria for inclusion and exclusion of studies

Parameter	Description
Participants	Healthy individuals and dyslipidemic individuals
Intervention/exposure	Krill oil supplementation administered for at least 2 wk
Comparison	Individuals supplemented with krill oil vs control individuals
Outcomes	Plasma/serum concentrations of at least one of the main lipid parameters (ie, total cholesterol, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, or triglycerides)
Study design	Randomized controlled trials with a parallel or crossover design

date, size of study population, trial design, dose and duration of intervention, control group allocation, baseline characteristics of study population (including age, gender, and body mass index), and changes in plasma concentrations of lipids were extracted.

Quality assessment

Risk of bias in the studies considered in this meta-analysis was evaluated according to the Cochrane instructions.¹⁵ Selection bias, performance bias, attrition bias, detection bias, reporting bias, and other sources of bias were judged to be high, low, or unclear in each of the studies included.

Quantitative data synthesis

Comprehensive Meta-Analysis V2 software (Biostat, NJ, USA)¹⁶ was used for statistical procedures. Plasma concentrations of lipids and lipoproteins were collated in milligram per deciliter. Inverse variance-weighted mean differences (WMDs) and 95% confidence limits were used as the summary statistics and calculated as previously described,^{17,18} considering a correlation coefficient of 0.5. When standard deviation values were not reported in a study, they were imputed by the pooled standard deviations from other studies.

Meta-analysis was performed under fixed- and random-effects models when I^2 (heterogeneity) values were <50% and $\geq 50\%$, respectively. For trials with more than one treatment arm and a single control group, the number of subjects in the control group was divided into a corresponding number of smaller groups. In order to evaluate the influence of each study on the overall effect size, sensitivity analysis was conducted using the one-study remove (leave-one-out) approach. Leave-one-out sensitivity analysis is a simple method to check if the overall estimate of effect is significantly driven by a single study or treatment arm. In this method, all included treatment arms are iteratively removed from the analysis, the effect size is calculated, and the significance of the pooled estimate is evaluated.¹⁹

Meta-regression

As potential confounders of treatment response, dosage and duration of supplementation with krill oil were entered into a meta-regression model to explore their association with the estimated effect size in each lipid species.

Publication bias

Funnel plot evaluation, Begg's rank correlation, and Egger's weighted regression tests were employed to assess the presence of publication bias in the meta-analysis. When there was evidence of funnel plot asymmetry, potentially missing studies were imputed using the trim-and-fill method. In the case of a significant result, the number of potentially missing studies required to make the *P* value nonsignificant was estimated using the fail-safe *N* method as another marker of publication bias.^{20–23}

RESULTS

Search results and trial flow

The initial screening for potential relevance removed the articles whose titles and/or abstracts were obviously irrelevant. Of the 15 full-text articles assessed for eligibility, 8 were excluded for the following reasons: did not measure plasma lipid levels (*n* = 4), investigated bioavailability of fatty acids only (*n* = 1), and did not have an appropriate randomized controlled trial [RCT] design (*n* = 3) (Figure 1).

After assessment, 7 RCTs (equivalent to 14 treatment arms) met the inclusion criteria and were used for the final meta-analysis. In total, 427 participants were allocated to a krill oil supplementation group and 235 to a control group in the selected studies.^{4,24–29} The number of participants in these trials ranged from 20 to 267. Included studies were published between 2004 and 2015 and were conducted in Canada (*n* = 3), Norway (*n* = 2), and the United States (*n* = 2). All studies used 500-mg capsules or softgels of krill oil, and the dosages ranged from 500 mg/d to 4 g/d. Duration of supplementation with krill oil ranged between 4 weeks and

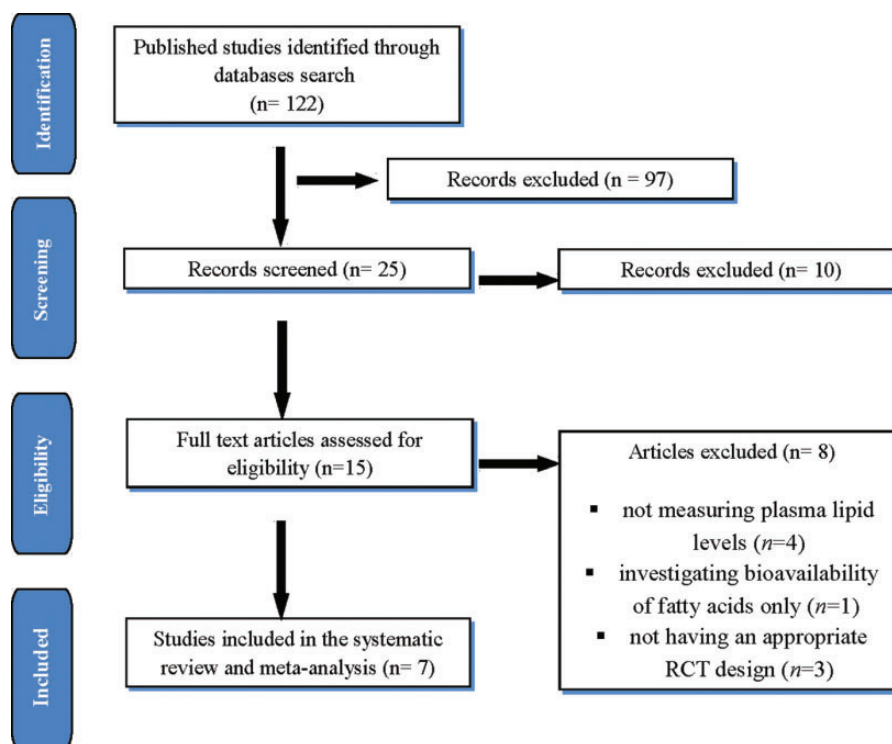


Figure 1 Flow diagram of the literature search process.

3 months. Four trials were designed as parallel-group studies and 3 had a crossover design. Three studies were multicenter. Demographic and baseline parameters of the included studies are shown in Table 2.^{4,24–29} The common side effects usually observed after administration of fish oil supplements, such as reflux, fishy aftertaste, or belching of fish flavors, were not reported after krill oil intake. Except for a few cases of mild to moderate gastrointestinal symptoms, krill oil was safe and well tolerated in all of the RCTs included in this review. No adverse events related to the supplementation were reported.

Risk-of-bias assessment

The risk of bias with respect to sequence generation and allocation concealment was unclear, but studies were of low risk in terms of other sources of bias. The systematic assessment of bias in the included studies is shown in Table 3.^{4,24–29}

Quantitative data synthesis

Total cholesterol. The meta-analysis of RCTs showed a reduction in plasma total cholesterol levels following supplementation with krill oil, but this reduction did not reach statistical significance (WMD, -7.5 mg/dL;

95%CI, -17.94 to 2.93 ; $P = 0.159$) (Figure 2). This effect size was robust in the sensitivity analysis (Figure 3). The effect of krill oil on plasma total cholesterol concentrations was not different in subgroups of RCTs with dosages of either <2 g/d or ≥ 2 g/d. With respect to treatment duration, there was a significant reduction in total cholesterol concentrations in the subset of RCTs lasting ≥ 12 weeks but not in the subset lasting <12 weeks (Table 4).

Visual inspection of the funnel plot as well as the results of Begg's rank correlation and Egger's linear regression tests suggested potential publication bias in the meta-analysis of krill oil's effects on plasma total cholesterol concentrations (Table 5 and Figure 4).

Low-density lipoprotein cholesterol. A significant reduction in plasma concentrations of LDL-C was observed following supplementation with krill oil (WMD, -15.52 mg/dL; 95%CI, -28.43 to -2.61 ; $P = 0.018$) (Figure 2). This effect size was sensitive to one of the included studies⁴ (Figure 3). The effect of krill oil on LDL-C concentrations did not reach statistical significance in subgroups of RCTs with dosages of either <2 g/d or ≥ 2 g/d. Regarding treatment duration, there was a significant reduction in plasma LDL-C in the subset of RCTs with supplementation durations ≥ 12 weeks

Table 2 Demographic characteristics and baseline parameters of the included studies

Study	Berge et al. (2014) ²⁴	Bunea et al. (2004) ⁴	Maki et al. (2009) ²⁵	Ramprasath et al. (2013) ²⁶	Ulven et al. (2011) ²⁷	Lobraico et al. (2015) ²⁸	Ramprasath et al. (2015) ²⁹
Location	Norway	Canada	USA	Canada	Norway	USA	Canada
Design	Multicenter randomized double-blind placebo-controlled parallel trial	Multicenter randomized double-blind placebo-controlled parallel trial	Multicenter randomized double-blind placebo-controlled parallel trial	Randomized double-blind placebo-controlled over trial	Randomized parallel group trial	Randomized double-blind placebo-controlled over trial	Randomized double-blind placebo-controlled over trial
Duration of trial	12 wk	3 mo	4 wk	4 wk	7 wk	4 wk	4 wk
Inclusion criteria	Subjects aged 21–79 y with a low habitual fatty fish and seafood intake (defined as intake of fatty fish and seafood ≤ 2 times per month) and borderline high or high fasting serum TG levels (defined as a fasting TG level of 150–499 mg/dL at screening visit, inclusive)	Patients aged 18–85 y with at least a 6-mo diagnosis of mildly high to very high blood cholesterol (193.9–347.9 mg/dL) and TG levels (203.8–354.4 mg/dL)	Healthy men and women aged 35–64 y with a waist circumference of ≥ 102 cm (men) or ≥ 88 cm (women)	Healthy volunteers	Healthy volunteers of both genders with normal or slightly elevated total blood cholesterol (< 7.5 mmol/L) and normal or slightly elevated blood triglyceride level (< 4.0 mmol/L)	Adults diagnosed with type 2 diabetes who were on oral glucose-lowering agents and/or insulin that could be confirmed by a healthcare provider	Healthy males and non-pregnant women aged 18–49 y
Krill oil form	500-mg capsule	500-mg softgel	500-mg capsule	500-mg capsule	500-mg capsule	500-mg capsule	500-mg capsule
Krill oil intervention	500 mg/d 1 g/d 2 g/d 4 g/d	1 g/d 1.5 g/d 2 g/d 3 g/d	2 g/d	3 g/d	3 g/d	1 g/d	500-mg capsule 1.5 g/d 3 g/d
No. of participants	Cases 53 53 51 58	23 7 19 11	25	24	36	47	20
Controls	52	30	25	37			

(continued)

Table 2 Continued

Study	Berge et al. (2014) ²⁴	Bunea et al. (2004) ⁴	Maki et al. (2009) ²⁵	Ramprasath et al. (2013) ²⁶	Ulven et al. (2011) ²⁷	Lobraico et al. (2015) ²⁸	Ramprasath et al. (2015) ²⁹
Location	Norway	Canada	USA	Canada	Norway	USA	Canada
Age (y) ^a	Cases 46.4 ± 12.7 40.5 ± 13.1 44.3 ± 12.1 45.5 ± 11.9	51 ± 9.46	49.4 ± 1.7 ^b	28.23 ± 5.35 ^b	40.3 ± 14.8	64.8 ± 8.8	NS
Male (%)	Controls Cases 66 64 73 67	NS	47.4 ± 1.6 ^b 12.0	50.0	40.5 ± 12.1 29.5	66.0	55.0
BMI (kg/m ²) ^a	Controls Cases 65 29.9 ± 3.3 29.0 ± 3.3 29.7 ± 2.6 29.3 ± 2.9	25.4 ± 3.9 (male) 28.2 ± 5.1 (female)	20.0 32.6 ± 1.5 ^b	23.76 ± 2.96 ^b	33.3 23.6 ± 3.3	30.5 ± 6.9	23.79 ± 0.71 ^b 23.62 ± 0.68 ^b
Total cholesterol (mg/dL) ^a	Controls Cases 221.1 ± 37.1 209.4 ± 40.3 221.4 ± 36.1 216.2 ± 31.2	235.83 ^c 231.19 ^c 247.42 ^c 250.52 ^c	33.3 ± 1.7 ^b 201.6 ± 6.1 ^b	168.85 ± 5.45 ^b	23.9 ± 3.0 192.61 ± 31.46	153.7 ± 32.6	23.61 ± 0.63 ^b 173.62 ± 7.06 ^b 168.02 ± 6.26 ^b
LDL-C (mg/dL) ^a	Controls Cases 207.9 ± 32.4 132.7 ± 4.6 120.8 ± 4.6 133.1 ± 4.7 125.8 ± 4.4	221.91 ^c 167.78 ^c 164.74 ^c 182.86 ^c 172.81 ^c	201.2 ± 7.3 ^b 129.7 ± 5.5 ^b	172.55 ± 5.69 ^b 93.05 ± 4.66 ^b	191.07 ± 35.70 118.50 ± 27.95	78.7 ± 28.4	169.56 ± 5.53 ^b 96.93 ± 7.25 ^b 94.28 ± 5.47 ^b
HDL-C (mg/dL) ^a	Controls Cases 44.5 ± 10.8 42.1 ± 10.6 42.1 ± 9.2 46.2 ± 10	57.22 ^c 58.76 ^c 51.03 ^c 64.18 ^c	125.5 ± 6.6 ^b 46.5 ± 2.8 ^b	95.03 ± 4.82 ^b 55.74 ± 2.91 ^b	115.03 ± 31.81 57.90 ± 14.20	44.6 ± 13.7	93.89 ± 5.07 ^b 55.48 ± 3.33 ^b 54.52 ± 3.14 ^b
TGs (mg/dL) ^a	Controls Cases 230.1 ± 71.4 238.9 ± 69.8 237.4 ± 73.1 220.2 ± 63	56.83 ^c 120.50 ^c 126.70 ^c 160.37 ^c 152.77 ^c	45.4 ± 2.1 ^b 126.9 ± 9.3 ^b	56.22 ± 3.38 ^b 101.10 ± 10.53 ^b	61.37 ± 13.66 97.35 ± 56.46	155.6 ± 95.3	54.71 ± 3.02 ^b 106.2 ± 9.62 ^b 98.02 ± 11.76 ^b
Controls	236.5 ± 80.9	143.53 ^c	152.8 ± 18.1 ^b	103.41 ± 9.84 ^b	81.42 ± 36.64		99.91 ± 10.02 ^b

Abbreviations: SD: standard deviation; SEM: standard error of the mean; BMI: body mass index; NS: not stated; LDL-C: low-density lipoprotein cholesterol; HDL-C: high-density lipoprotein cholesterol; TG, triglyceride.

^aValues are expressed as mean ± SD unless otherwise indicated.

^bValues are expressed as mean ± SEM.

^cMean value.

Table 3 Risk of bias in the included studies as assessed using Cochrane criteria

Reference	Sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other potential threats to validity
Berge et al. (2014) ²⁴	Unclear	Unclear	Low	Low	Low	Low	Low
Bunea et al. (2004) ⁴	Unclear	Unclear	Low	Low	Low	Low	Low
Maki et al. (2009) ²⁵	Unclear	Unclear	Low	Low	Low	Low	Low
Ramprasath et al. (2013) ²⁶	Unclear	Unclear	Low	Low	Low	Low	Low
Ulven et al. (2011) ²⁷	Unclear	Unclear	Low	Low	Low	Low	Low
Lobraico et al. (2015) ²⁸	Low	Low	Low	Low	Low	Low	Low
Ramprasath et al (2015) ²⁹	Unclear	Unclear	Low	Low	Low	Low	Low

but not in the subset with supplementation lasting <12 weeks (Table 4).

Visual inspection of the funnel plot as well as the results of Begg's rank correlation and Egger's linear regression tests suggested potential publication bias in the meta-analysis of krill oil's effects on plasma LDL-C concentrations (Table 4 and Figure 4).

High-density lipoprotein cholesterol. The meta-analysis of RCT data suggested a significant elevation in plasma HDL-C concentrations following krill oil supplementation (WMD, 6.65 mg/dL; 95%CI, 2.30–10.99; $P=0.003$) (Figure 2). This effect size was robust in the sensitivity analysis, and iterative removal of each single study from the meta-analysis did not affect the statistical significance of the effect size (Figure 3). In the subgroup analysis, there was a significant increase in plasma HDL-C concentrations in the subset of trials with krill oil dosages of ≥ 2 g/d but not in the subset with dosages of <2 g/d. With respect to the duration of supplementation, HDL-C elevations remained significant in both study subsets with supplementation durations of <12 weeks and ≥ 12 weeks, though a greater reduction was observed in the latter subgroup (Table 4).

Visual inspection of the funnel plot as well as the results of Begg's rank correlation and Egger's linear regression tests suggested potential publication bias in the meta-analysis of krill oil's effects on plasma HDL-C concentrations (Table 5 and Figure 4).

Triglycerides. A significant reduction in plasma TG concentrations following krill oil supplementation was found in the meta-analysis of RCTs (WMD, -14.03 mg/dL; 95%CI, -21.38 to -6.67 ; $P<0.001$) (Figure 2). This effect size was robust in the sensitivity analysis, and iterative removal of each single study from the meta-analysis did not affect the statistical significance of the effect size (Figure 3). In the subgroup analysis, there was a significant reduction in plasma TG concentrations in both subsets of trials with krill oil doses of <2 g/d and ≥ 2 g/d. However, with respect to supplementation duration, significant reduction of plasma TG concentrations was observed only in the

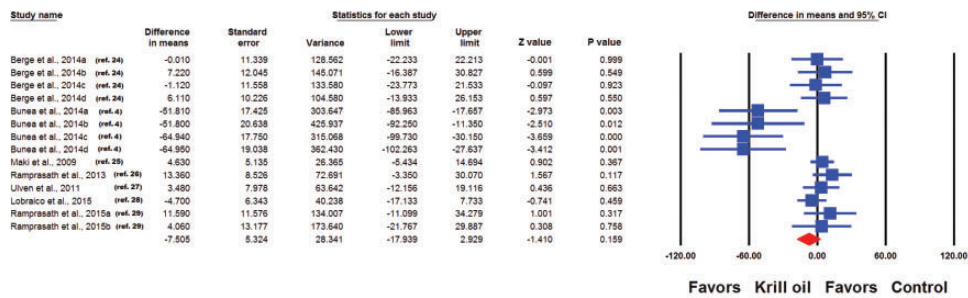
subset of RCTs lasting ≥ 12 weeks and not in the subset lasting <12 weeks (Table 4). Although the results of Begg's rank correlation and Egger's linear regression tests did not indicate any publication bias (Table 5), the funnel plot was slightly asymmetric, suggesting potential publication bias in the meta-analysis of krill oil's effects on plasma TG concentrations. Using Duval and Tweedie trim-and-fill correction, 3 potentially missing studies were imputed on the left side of funnel plot, yielding a corrected effect size of -15.12 mg/dL (95%CI, -22.00 to -8.23) (Figure 4).

DISCUSSION

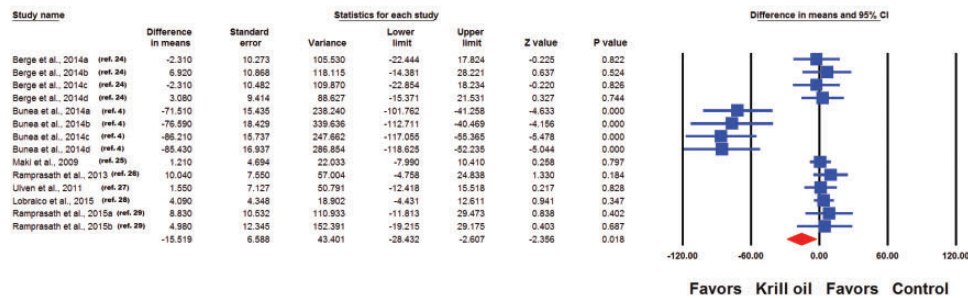
To the best of knowledge, the current systematic review and meta-analysis is the first to analyze evidence from RCTs on the effect of krill oil supplementation on plasma lipids. The results showed a significant reduction in plasma concentrations of LDL-C and TGs and a significant elevation in plasma HDL-C concentrations following supplementation with krill oil, while a reduction in plasma total cholesterol levels did not reach statistical significance. When the studies were stratified according to their duration, there was a significant reduction in total cholesterol, LDL-C, and TGs in the subset of RCTs lasting ≥ 12 weeks but not in the subset lasting <12 weeks, while HDL-C elevations remained significant in both study subsets.

The favorable effects of krill oil on the lipid profile found in the current meta-analysis suggest the use of this marine product as an interesting add-on therapy in dyslipidemic patients. However, the exact mechanisms responsible for the lipid-lowering effects of krill oil are not completely understood. It has been shown that krill oil acts more as an antihyperlipidemic product than a hypolipidemic one, since TGs and cholesterol levels are lowered in patients with elevated blood lipids, but not in healthy volunteers.²⁶ It has been proved that krill oil reduces the expression of genes required for lipid and isoprenoid/cholesterol synthesis.³⁰ In contrast, fish oil modulates expression of several genes, such as peroxisome proliferator-activated receptor (PPAR)- α , sterol regulatory element-binding proteins (SREBP)-1, and

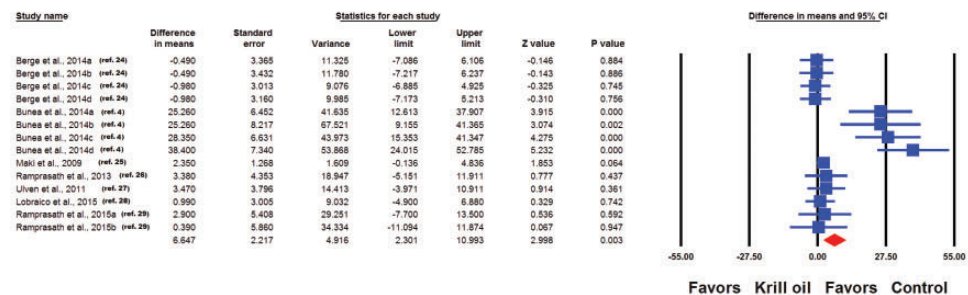
TC



LDL-C



HDL-C



TG

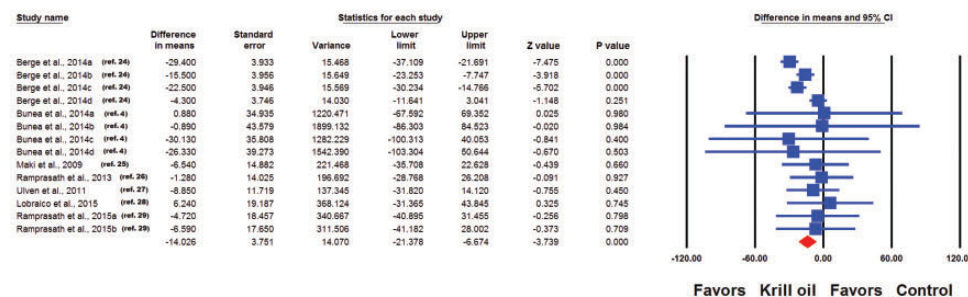
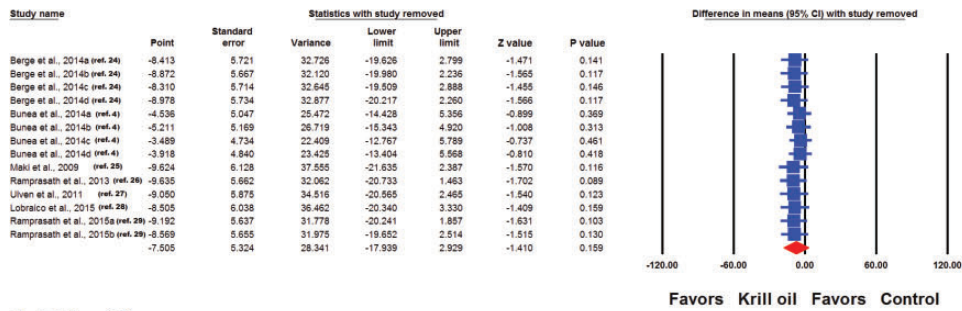
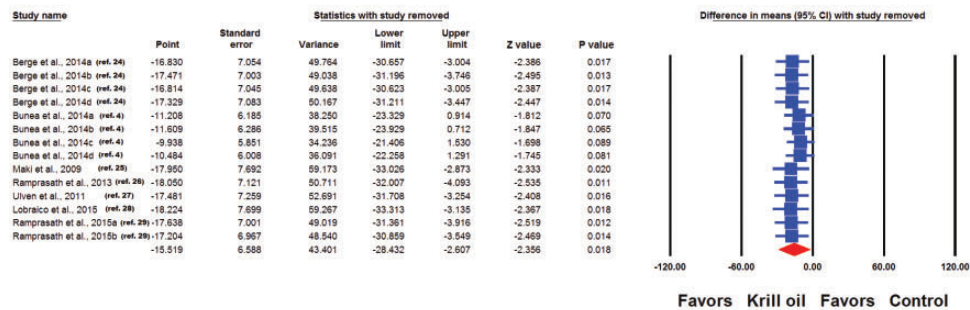


Figure 2 Forest plots detailing weighted mean differences and 95% confidence intervals for the effect of krill oil supplementation on plasma lipid concentrations. Abbreviations: HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; TG, triglycerides.

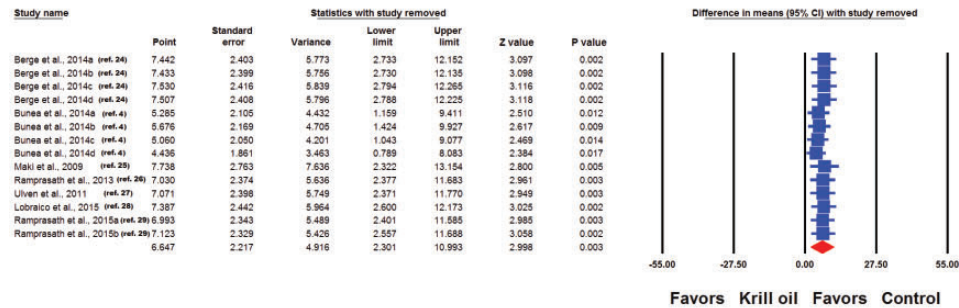
TC



LDL-C



HDL-C



TG

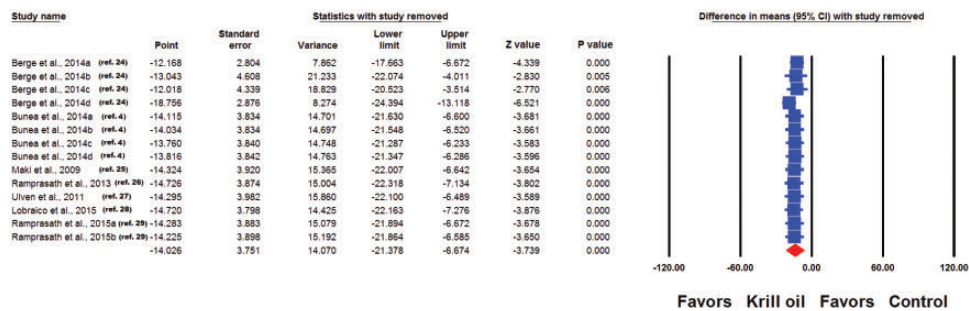


Figure 3 Leave-one-out sensitivity analysis for the effects of krill oil supplementation on plasma lipid concentrations. Abbreviations: HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; TG, triglycerides.

Table 4 Meta-analysis of the effects of krill oil on plasma lipid concentrations in the subgroups of studies with different dosages and durations of supplementation

	Total cholesterol (mg/dL)		LDL-C (mg/dL)		HDL-C (mg/dL)		Triglycerides (mg/dL)	
	WMD (95%CI)	P value	WMD (95%CI)	P value	WMD (95%CI)	P value	WMD (95%CI)	P value
Dosage <2 g/d	-11.12 (-27.53 to 5.29)	0.184	-18.85 (-42.32 to 4.61)	0.115	6.44 (-1.06 to 13.95)	0.092	-21.31 (-26.63 to -16.00)	<0.001
Dosage ≥2 g/d	-5.55 (-19.80 to 8.69)	0.445	-14.33 (-31.98 to 3.32)	0.112	7.25 (1.27 to 13.24)	0.018	-12.16 (-17.11 to -7.21)	<0.001
Duration <12 wk	3.86 (-2.17 to 9.89)	0.210	3.91 (-1.14 to 8.95)	0.129	2.29 (0.24 to 4.33)	0.028	-4.60 (-16.82 to 7.61)	0.460
Duration ≥12 wk	-24.12 (-45.37 to -2.86)	0.026	-37.14 (-64.96 to -9.31)	0.009	12.53 (3.44 to 21.63)	0.007	-17.67 (-27.15 to -8.19)	<0.001

Abbreviations: LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; WMD, weighted mean difference.

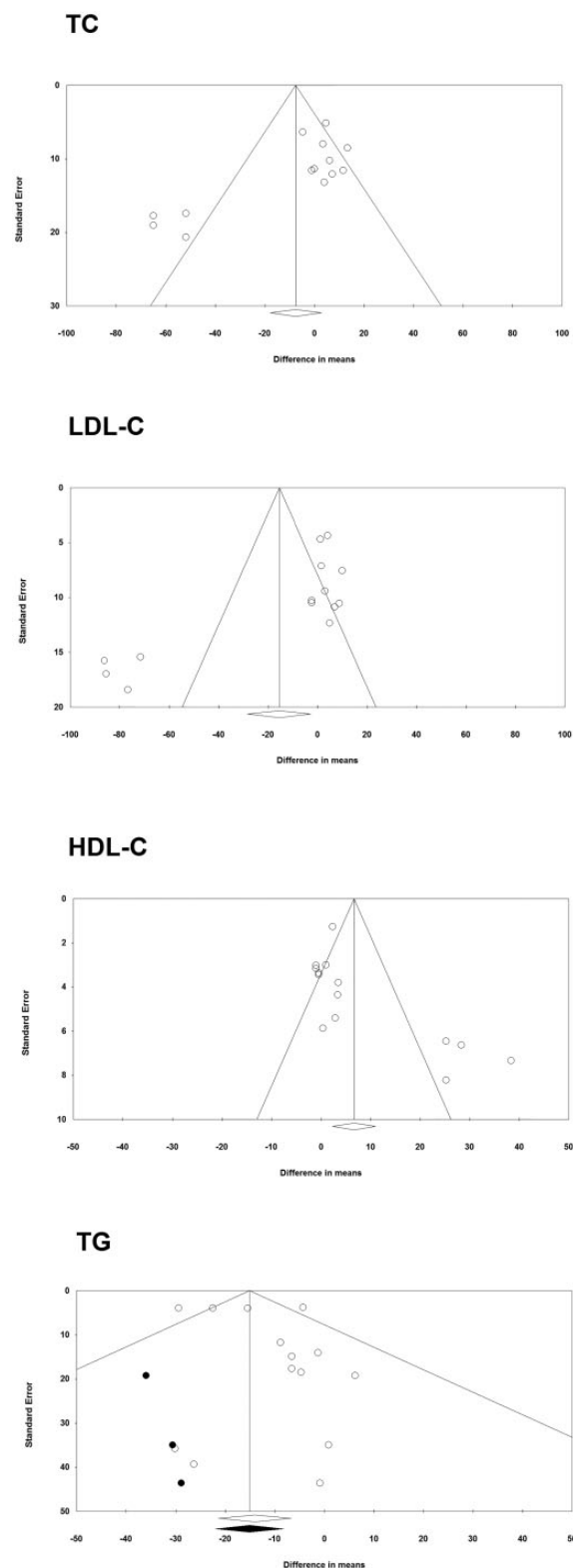


Figure 4 Funnel plots detailing publication bias in the studies selected for analysis. Trim-and-fill method was used to impute for potentially missing studies. Open circles represent observed published studies; closed circles represent imputed unpublished studies. Abbreviations: HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; TG, triglycerides.

Table 5 Publication bias in the meta-analysis of studies reporting the effects of krill oil supplementation on plasma lipids

Plasma lipid	Begg's rank correlation test			Egger's linear regression test				
	Kendall's tau ^a	z value	P value ^b	Intercept (95%CI)	t	df	P value ^b	No. of studies ^c
Total cholesterol	−0.46	2.30	0.021	−3.00 (−5.20 to −0.80)	3.82	9	0.004	N/A
LDL-C	−0.55	2.74	0.006	−4.15 (−6.95 to −1.34)	3.22	12	0.007	44
HDL-C	0.59	2.96	0.003	2.35 (0.21 to 4.49)	2.40	12	0.034	91
Triglycerides	−0.07	0.33	0.743	0.59 (−0.72 to 1.89)	0.98	12	0.346	105

Abbreviations: HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol.

^aWith continuity correction.

^bTwo-tailed.

^cNumber of studies (calculated using the fail-safe N method) required to make the P value nonsignificant.

(SREBP)-2, all involved in triacylglycerol and fatty acid catabolism in skeletal muscle and liver.³¹

A higher bioavailability of long-chain n-3 polyunsaturated fatty acids and a higher therapeutic effect¹ has been observed in studies comparing the efficacy of krill oil with that of fish oil.^{5,26,27,32} The association between long-chain n-3 fatty acids and phospholipids is responsible for lipogenic effects of krill oil, while the association between n-3 fatty acids and TGs is culpable for the lipogenic effects of fish oil.³ Phospholipids are amphipathic molecules that do not require emulsification via bile salts like the hydrophobic TGs, hence they are able to move easily across the intestinal wall and cell membranes.⁵ Therefore, the association between long-chain n-3 fatty acids and phospholipids might have an important role in improving the bioavailability and absorption of fatty acids through the intestinal wall and enhancing the n-3:n-6 ratio.²⁷ The different effects of krill oil as compared with fish oil on lipid parameters in experimental studies might be also explained by the different effects of these oils on the secretion of very low-density lipoprotein and the distinct modulation of gene expression in the intestine and the liver.³³ Moreover, experimental studies showed that krill oil is capable of upregulating the activity of the mitochondrial respiratory chain and downregulating the activity of pathways involved in lipid and cholesterol synthesis and hepatic glucose production, while fish oil did not modulate the same metabolic pathways regulated by krill oil.³⁰

While fish oil upregulates the cholesterol synthesis pathway, the effect of krill oil is the opposite.³⁰ An experimental study in rats fed a krill-oil-enriched diet has shown that krill oil might have the capacity to inhibit de novo lipogenesis by decreasing the activity of both cytosolic acetyl coenzyme A carboxylase and the mitochondrial citrate carrier and to increase the oxidation of fatty acids by decreasing the fatty acid synthetase activity.^{34,35} Another unique feature of krill oil is related to its powerful antioxidant astaxanthin, which has a slight, but not significant, glucose-lowering effect on plasma lipid concentrations.³⁶ Another important antioxidant found in krill oil is vitamin E, known for its capacity to protect biological membranes against lipid

peroxidation.³⁷ Nevertheless, krill oil is a rich source of high-quality protein in the range of 60% to 65% dry weight, including all 9 essential amino acids required by adults.³⁸

One study showed that daily consumption of 3 g of krill oil formulated with 543 mg of EPA and DHA raises the plasma EPA and DHA concentrations exactly the same as intake of fish oil containing 864 mg of EPA and DHA.²⁷ Another new study in healthy subjects showed that the proportion of EPA and DHA in plasma phospholipid fatty acids measured during a 72-hour follow-up as an incremental area under the curve was significantly larger after krill oil ingestion than after fish oil ingestion.⁹ The adverse effects observed after krill oil consumption are observed only in individuals with known allergies to different crustaceans or shellfish, such as shrimp or crabs.^{1,39} Moreover, patients who take krill oil and anticoagulant/antiplatelet drugs simultaneously might have an increased risk of bleeding.

The present meta-analysis has some limitations. Most of the included studies had a small number of participants and were heterogeneous with regard to the characteristics of patients. Moreover, the lipid profile and content of krill oil could fluctuate appreciably according to conditions such as the krill species used, the age of the krill, the season harvested, and the delay time between collection and freezing. The effects of krill on small, dense LDL subfractions and HDL function do not seem to have been investigated. The function of HDL may be even more relevant than its measured levels.¹³ However, the effect of krill oil on TG and HDL-C levels suggests there will be an improvement in small, dense LDL subfractions.⁴⁰ The baseline dietary data and the prescribed diet were not clearly described in all the studies included. In this meta-analysis, the most significant effects on plasma lipids were reported in the treatment arms in the study by Bunea et al.⁴ Although the impact of each treatment arm on the overall estimated effect size was explored in the sensitivity analysis, another attempt was made to assess the impact of this study on the effect size by removing all treatment arms in this study from the meta-analysis. The results revealed that, although the TG-lowering effect of krill oil remains significant, the effects on LDL-C and

HDL-C were no longer evident when the study of Bunea et al.⁴ was excluded. This highlights the need for future studies to better evaluate whether krill oil supplementation exerts a LDL-lowering effect. Finally, the value of adding krill oil to conventional and novel lipid-lowering medications^{41–44} remains unknown.

CONCLUSION

The present meta-analysis of RCTs suggests the efficacy of krill oil supplementation in lowering plasma concentrations of LDL-C and TGs as well as increasing those of HDL-C. Further, well-designed clinical studies with higher numbers of participants are necessary to assess the impact of krill oil supplementation on other indices of cardiometabolic risk and on the risk of cardiovascular outcomes. The value of adding krill oil to statins and other routinely administered lipid-lowering therapies is also open to question.

Acknowledgments

Funding/support. No external funding was received for this work.

Declaration of interest. The authors have no relevant interests to declare.

REFERENCES

1. Kwantes JM, Grundmann O. A brief review of krill oil history, research, and the commercial market. *J Diet Suppl.* 2015;12:23–35.
2. Deutsch L. Evaluation of the effect of Neptune Krill Oil on chronic inflammation and arthritic symptoms. *J Am Coll Nutr.* 2007;26:39–48.
3. Burri L, Johnsen L. Krill products: an overview of animal studies. *Nutrients.* 2015;7:3300–3321.
4. Bunea R, El Farrah K, Deutsch L. Evaluation of the effects of Neptune Krill Oil on the clinical course of hyperlipidemia. *Altern Med Rev.* 2004;9:420–428.
5. Sampalis F, Bunea R, Pelland MF, et al. Evaluation of the effects of Neptune Krill Oil on the management of premenstrual syndrome and dysmenorrhea. *Alternative Med Rev.* 2003;8:171–179.
6. Xiao SU, Tanalgo P, Bustos M, et al. The effect of krill oil and n-3 polyunsaturated fatty acids on human osteosarcoma cell proliferation and migration. *Curr Drug Targets.* 2015;16. doi:10.2174/1389450116666150107152206.
7. Ierna M, Kerr A, Scales H, et al. Supplementation of diet with krill oil protects against experimental rheumatoid arthritis. *BMC Musculoskelet Disord.* 2010;11:136. doi:10.1186/1471-2474-11-136.
8. Salem N Jr, Kuratko CN. A reexamination of krill oil bioavailability studies. *Lipids Health Dis.* 2014;13:137. doi:10.1186/1476-511X-13-137.
9. Köhler A, Sarkkinen E, Tapola N, et al. Bioavailability of fatty acids from krill oil, krill meal and fish oil in healthy subjects – a randomized, single-dose, cross-over trial. *Lipids Health Dis.* 2015;14:19. doi:10.1186/s12944-015-0015-4.
10. Cicero AFG, Colletti A. Krill oil: evidence of a new source of polyunsaturated fatty acids with high bioavailability. *Clin Lipidol.* 2015;10:1–4.
11. Krill oil monograph. *Altern Med Rev.* 2010;15:84–86.
12. Kris-Etherton PM, Fleming JA. Emerging nutrition science on fatty acids and cardiovascular disease: nutritionists' perspectives. *Adv Nutr.* 2015;6(suppl): 326S–337S.
13. Otsuka-Kmieciak A, Mikhailidis DP, Nicholls SJ, et al. Dysfunctional HDL: a novel important diagnostic and therapeutic target in cardiovascular disease? *Prog Lipid Res.* 2012;51:314–324.

14. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ.* 2009;339:b2535. doi:10.1136/bmj.b2535.
15. Higgins JPT, Green S, eds. *Cochrane Handbook for Systematic Reviews of Interventions.* Version 5.1.0 [updated March 2011]. London, England: The Cochrane Collaboration; 2011.
16. Borenstein M, Hedges L, Higgins J, et al. *Comprehensive Meta-analysis.* Version 2. Englewood, NJ: Biostat; 2005:104.
17. Michalska-Kasiczak M, Sahebkar A, Mikhailidis DP, et al. Analysis of vitamin D levels in patients with and without statin-associated myalgia – a systematic review and meta-analysis of 7 studies with 2420 patients. *Int J Cardiol.* 2015;178:111–116.
18. Ferretti G, Bacchetti T, Sahebkar A. Effect of statin therapy on paraoxonase-1 status: a systematic review and meta-analysis of 25 clinical trials. *Prog Lipid Res.* 2015;60:50–73.
19. Greenhouse JB, Iyengar S. Sensitivity analyses and diagnostics. In: Cooper H, Hedges LV, Valentine JC, eds. *The Handbook of Research Synthesis.* New York, NY: Russell Sage Foundation; 1994:383–398.
20. Sahebkar A. Does *PPARG* gene Pro12Ala polymorphism affect nonalcoholic fatty liver disease risk? Evidence from a meta-analysis. *DNA Cell Biol.* 2013;32:188–198.
21. Banach M, Serban C, Sahebkar A, et al. Impact of statin therapy on coronary plaque composition: a systematic review and meta-analysis of virtual histology intravascular ultrasound studies. *BMC Med.* 2015;13:229. doi:10.1186/s12916-015-0459-4.
22. Duval S, Tweedie R. Trim and fill: a simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics.* 2000;56:455–463.
23. Duval S, Tweedie R. A nonparametric “trim and fill” method of accounting for publication bias in meta-analysis. *J Am Stats Assoc.* 2000;95:89–98.
24. Berge K, Musa-Veloso K, Harwood M, et al. Krill oil supplementation lowers serum triglycerides without increasing low-density lipoprotein cholesterol in adults with borderline high or high triglyceride levels. *Nutr Res.* 2014;34:126–133.
25. Maki KC, Reeves MS, Farmer M, et al. Krill oil supplementation increases plasma concentrations of eicosapentaenoic and docosahexaenoic acids in overweight and obese men and women. *Nutr Res.* 2009;29:609–615.
26. Ramprasath VR, Eyal I, Zchut S, et al. Enhanced increase of omega-3 index in healthy individuals with response to 4-week n-3 fatty acid supplementation from krill oil versus fish oil. *Lipids Health Dis.* 2013;12:178. doi:10.1186/1476-511X-12-178.
27. Ulven SM, Kirkhus B, Lamglait A, et al. Metabolic effects of krill oil are essentially similar to those of fish oil but at lower dose of EPA and DHA, in healthy volunteers. *Lipids.* 2011;46:37–46.
28. Lobraico JM, DiLello LC, Butler AD, et al. Effects of krill oil on endothelial function and other cardiovascular risk factors in participants with type 2 diabetes, a randomized controlled trial. *BMJ Open Diab Res Care.* 2015;3:e000107. doi:10.1136/bmjdc-2015-000107.
29. Ramprasath VR, Eyal I, Zchut S, et al. Supplementation of krill oil with high phospholipid content increases sum of EPA and DHA in erythrocytes compared with low phospholipid krill oil. *Lipids Health Dis.* 2015;14:142. doi:10.1186/s12944-015-0142-y.
30. Burri L, Berge K, Wibrand K, et al. Differential effects of krill oil and fish oil on the hepatic transcriptome in mice. *Front Genet.* 2011;2:45. doi:10.3389/fgene.2011.00045.
31. Gessner DK, Grone B, Rosenbaum S, et al. Effect of dietary fish oil on the expression of genes involved in lipid metabolism in liver and skeletal muscle of lactating sows. *J Anim Physiol Anim Nutr (Berlin).* 2016;100:337–347.
32. Salem N Jr, Kuratko CN. A reexamination of krill oil bioavailability studies. *Lipids Health Dis.* 2014;13:137. doi:10.1186/1476-511X-13-137.
33. Tillander V, Bjørndal B, Burri L, et al. Fish oil and krill oil supplementations differentially regulate lipid catabolic and synthetic pathways in mice. *Nutr Metab (London).* 2014;11:20. doi:10.1186/1743-7075-11-20.
34. Ferramosca A, Conte L, Zara V. A krill oil supplemented diet reduces the activities of the mitochondrial tricarboxylate carrier and of the cytosolic lipogenic enzymes in rats. *J Anim Physiol Anim Nutr (Berlin).* 2012;96:295–306.
35. Ferramosca A, Conte A, Burri L, et al. A krill oil supplemented diet suppresses hepatic steatosis in high-fat fed rats. *PLoS One.* 2012;7:e38797. doi:10.1371/journal.pone.0038797.
36. Ursoniu S, Sahebkar A, Serban M-C, et al. Lipid profile and glucose changes after supplementation with astaxanthin: a systematic review and meta-analysis of randomized controlled trials. *Arch Med Sci.* 2015;11:253–266.
37. Tou JC, Jaczynski J, Chen YC. Krill for human consumption: nutritional value and potential health benefits. *Nutr Rev.* 2007;65:63–77.
38. Chen YC, Tou J, Jaczynski J. Amino acid and mineral composition of protein and other components and their recovery yields from whole Antarctic krill (*Euphausia superba*) using isoelectric solubilization/precipitation. *J Food Sci.* 2009; 74:H31–H39.
39. Bengtson Nash SM, Schlabach M, Nichols PD. A nutritional-toxicological assessment of Antarctic krill oil versus fish oil dietary supplements. *Nutrients.* 2014;6:3382–3402.

40. Mikhailidis DP, Elisaf M, Rizzo M, et al. "European panel on low density lipoprotein (LDL) subclasses": a statement on the pathophysiology, atherogenicity and clinical significance of LDL subclasses. *Curr Vasc Pharmacol*. 2011;9:531–532.
41. Sahebkar A, Chew GT, Watts GF. Recent advances in pharmacotherapy for hypertriglyceridemia. *Prog Lipid Res*. 2014;56:47–66.
42. Sahebkar A, Watts GF. New LDL-cholesterol lowering therapies: pharmacology, clinical trials, and relevance to acute coronary syndromes. *Clin Ther*. 2013;35:1082–1098.
43. Dragan S, Serban MC, Banach M. Proprotein convertase subtilisin/kexin 9 inhibitors: an emerging lipid-lowering therapy? *J Cardiovasc Pharmacol Ther*. 2015;20:157–168.
44. Banach M, Aronow WS, Serban C, et al. Lipids, blood pressure and kidney update 2014. *Pharmacol Res*. 2015;95–96:111–125.